

BLDC Motor Failure Detection System FOR DRONE SAFETY

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Drones are widely used across several fields, including aerial photography, surveying, inspection, agriculture, and recreation. Despite their growing importance, many drones develop hardware faults such as motor disconnection, ESC faults, missing propellers, over-current conditions, or low battery voltage before take-off. These issues can cause sudden imbalance, crashes, costly component damage, and loss of user confidence.

Apart from software diagnostics or GPS-based checks, drone controllers typically perform only basic system checks. As a result, critical faults often remain undetected. This system introduces a smart drone safety mechanism that functions as a pre-flight safety guardian. It monitors motor current and battery voltage using a precision current sensor and a voltage divider network. With a single button press, the ESC powers the motor for a

short, controlled test cycle while the system analyses real-time current behaviour and battery health.

Very low current may indicate ESC disconnection or

non-responsiveness. Low current may also suggest loose motor wiring or partial disconnection. Mid-range current indicates that the motor is operating without a propeller.

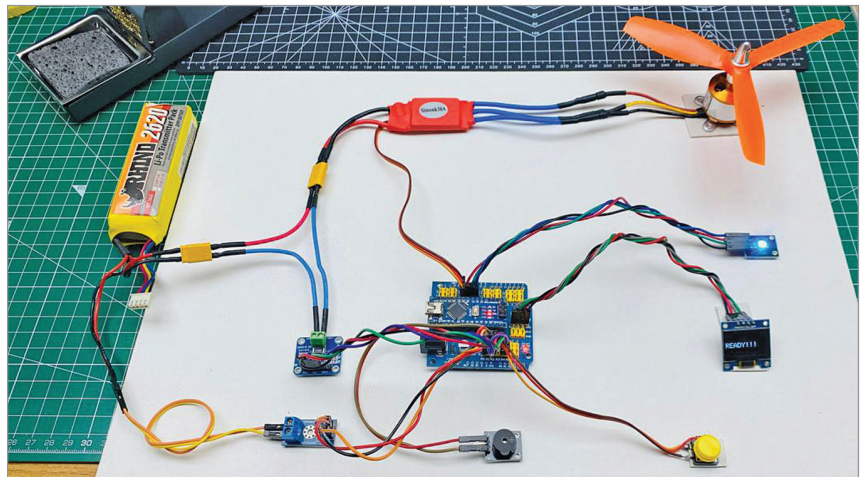


Fig. 1: Author's prototype

TABLE 1
BILL OF MATERIALS

Components	Quantity
Arduino Nano (MOD1)	1
Arduino Nano expansion board	1
2.4cm (0.96") I ² C OLED display (OLED)	1
Buzzer module (B1)	1
INA219 current sensor module (S1)	1
Voltage sensor module (S2)	1
Electronic speed controller (ESC)	1
BLDC motor (M1)	1
RGB LED module (U2)	1
Push button switch (SW1)	1
220Ω resistor (R1)	1
10kΩ resistor (R2)	1
Jumper wires (male-male/mixed)	As required
12V Li-Po drone battery (U1)	1

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1
2 #include <Wire.h>
3 #include <Adafruit_INA219.h>
4 #include <Adafruit_GFX.h>
5 #include <Adafruit_SSD1306.h>
6 #include <Servo.h>
7
8 // Display
9 #define SCREEN_WIDTH 128
10 #define SCREEN_HEIGHT 64
11 #define OLED_RESET -1
12 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
13
14 // Sensors & ESC
15 Adafruit_INA219 ina219;
16 Servo esc;
17
18 // Pins
19 #define TEST_BUTTON A2
20 #define ESC_PIN 11
21 #define BUZZER_PIN A1
22 #define RGB_RED 10
23 #define RGB_GREEN 9
24 #define RGB_BLUE 8
25 #define VOLTAGE_SENSOR A0
26
27 // THRESHOLDS - Simple logical operators
28 #define NO_ESC_THRESHOLD 10 // < 10mA = NO ESC error
29 #define NO_MOTOR_MIN 70 // > 70mA = NO Motor error start
30 #define NO_MOTOR_MAX 200 // < 200mA = NO Motor error end
31 #define NO_MOTOR_WIRE_THRESHOLD 300 // < 300mA while testing = NO Motor wire (moi
32 #define NO_PROP_MIN 400 // > 400mA = No Propeller error start

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Fig. 2: Code snippet of the source code